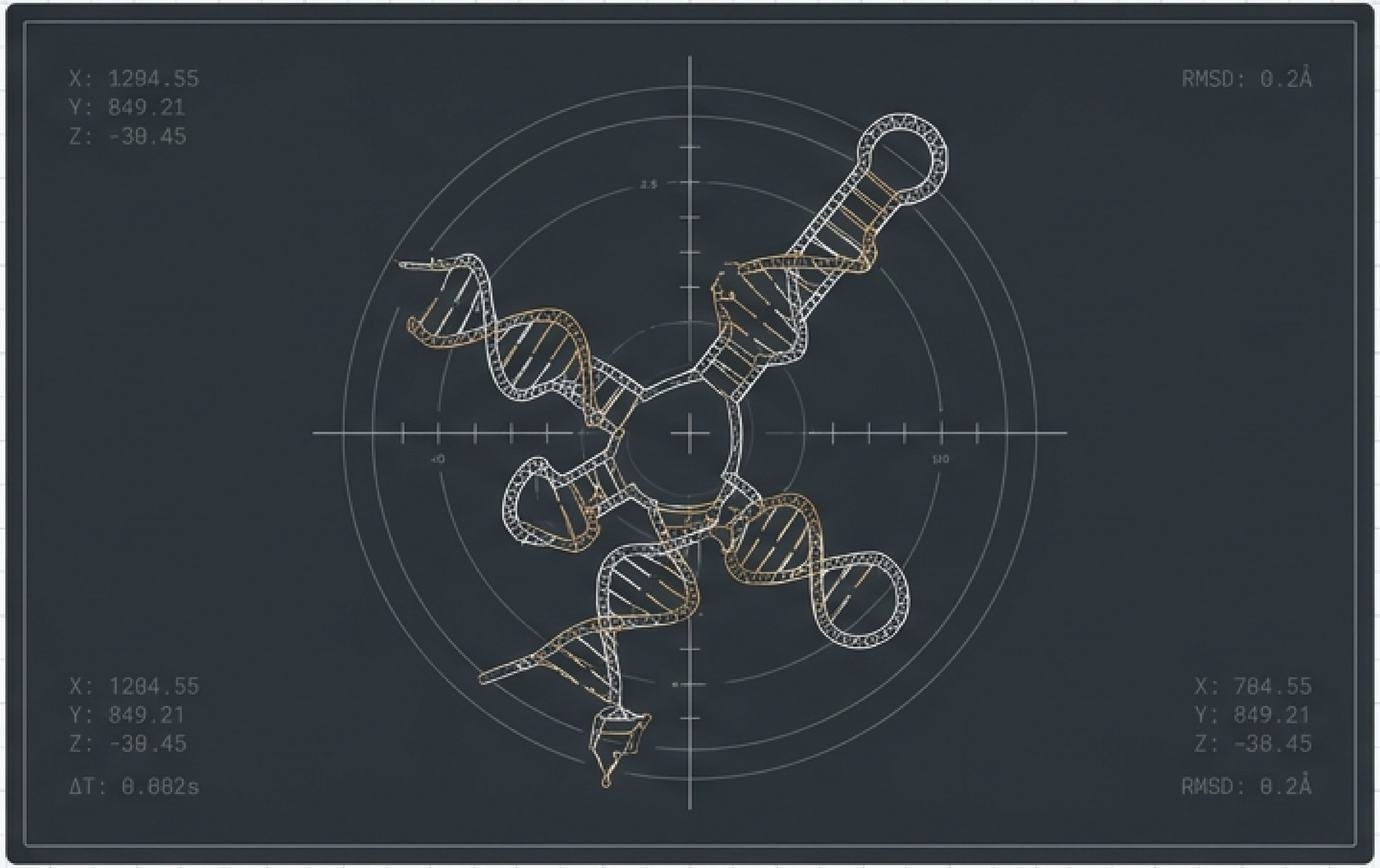
ENGINE: v6.0
COARSE-GRAINED

STATUS:
PRODUCTION-SAFE



SynthForge-RNA3D: A Multimodal Ground Truth for Bio-AI

ML-grade synthetic structural biology for geometric deep learning, diffusion models, and cross-modal alignment.

LICENSE: MIT

COORD_SYS: CARTESIAN_3D

The Physics-Informed Bridge for Machine Learning

	Atomistic MD (e.g., OpenMM)	SynthForge-RNA3D (v6.0)	Static Prediction (e.g., AlphaFold3)
Speed & Throughput	Days/Weeks	Minutes (High Throughput)	Seconds
Physical Realism	Full all-atom quantum/classical mechanics	Coarse-grained stochastic thermodynamics	Learned spatial approximation (no physics engine)
Output Type	Atom-level Trajectory	Time-resolved Multi-modal Trajectory (1D to 4D)	Single Static Endpoint
Primary ML Use-Case	Micro-state validation	Large-scale GNN/Diffusion pretraining	Fast inference target

Deep Dive

Scope Clarification: SynthForge v6.0 is benchmark-quality and ML-grade.

It is not an atomistic MD atomistic MD replacement, but a scalable proxy environment for structural representation learning.

The Physics-Inspired Simulation Engine (v6.0)

1. Stochastic Dynamics

Langevin-style dynamics with thermal noise injection.

2. Ionic Environment

Debye-Hückel ionic modeling and electrostatic screening.

3. Thermodynamic Modeling

Simulated annealing accounting for ΔG, ΔH, and ΔS.

4. Structural Constraints

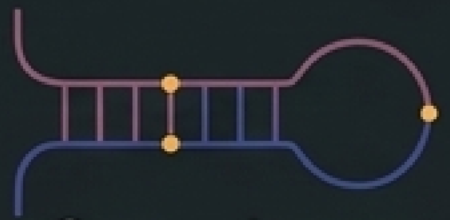
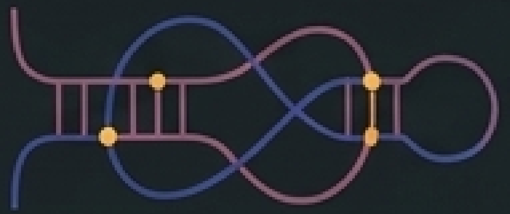
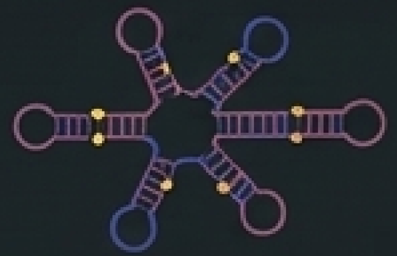
Dihedral regularization, angle potentials, Lennard-Jones interactions, and hydrogen bonding.

Stability Enhancements: Proper exclusion masks and stable nonbonded handling.

Deep Dive:

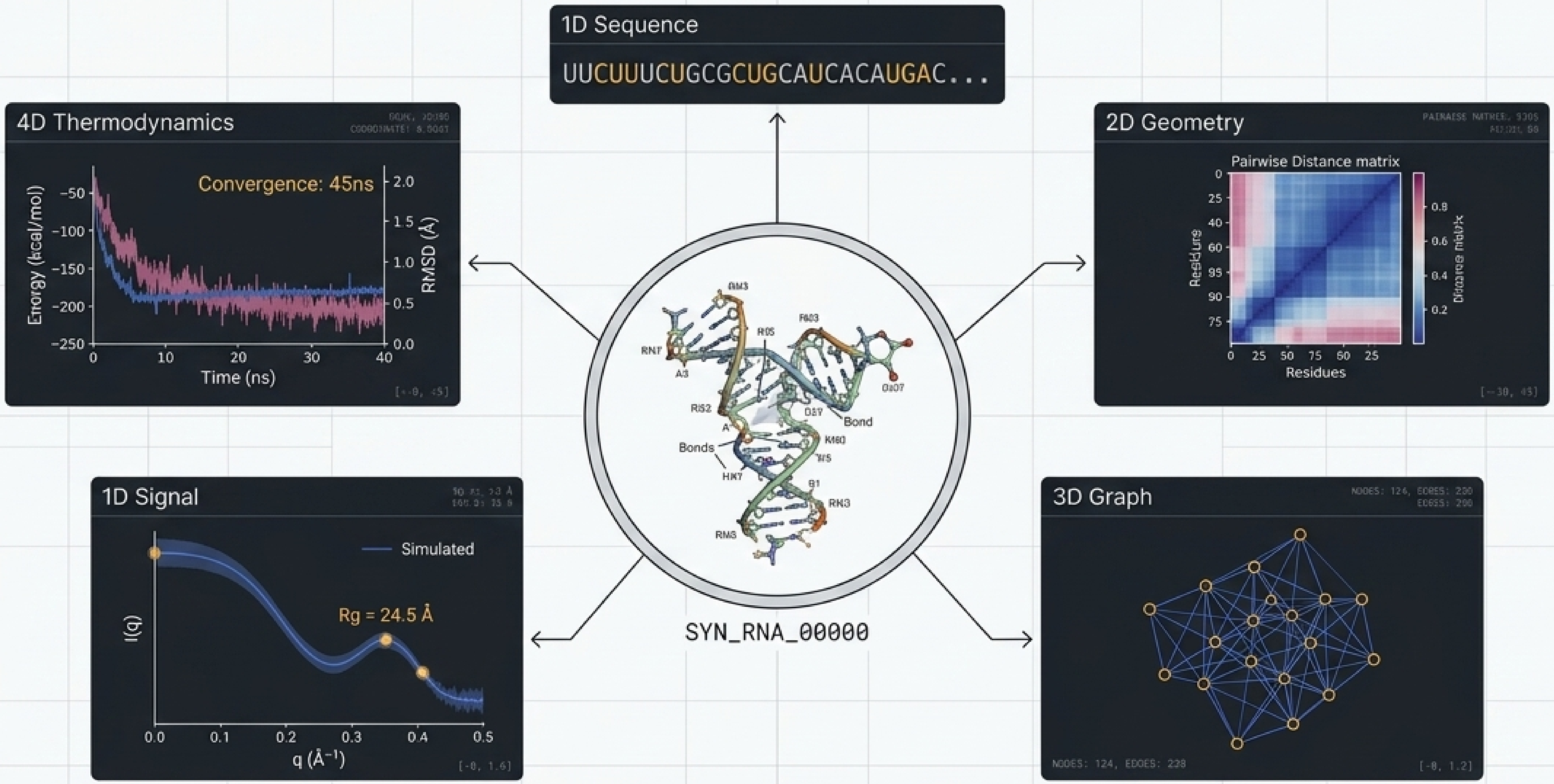
Langevin Dynamics: By injecting controlled thermal noise, the engine forces the RNA chain to escape local energy minima, producing biologically relevant folding pathways rather than instantaneous collapse.

Configurable Topological Diversity

 Stem-loop	**Composition:** Standard contiguous native base-pairing.	**ML Challenge:** Baseline structural motif recognition.
 Pseudoknot	**Composition:** Interleaved native base-pairs.	**ML Challenge:** 3D spatial crossing and non-planar graph representation.
 Multi-junction	**Composition:** Complex branching architectures.	**ML Challenge:** Global coordinate alignment and structural stability prediction.

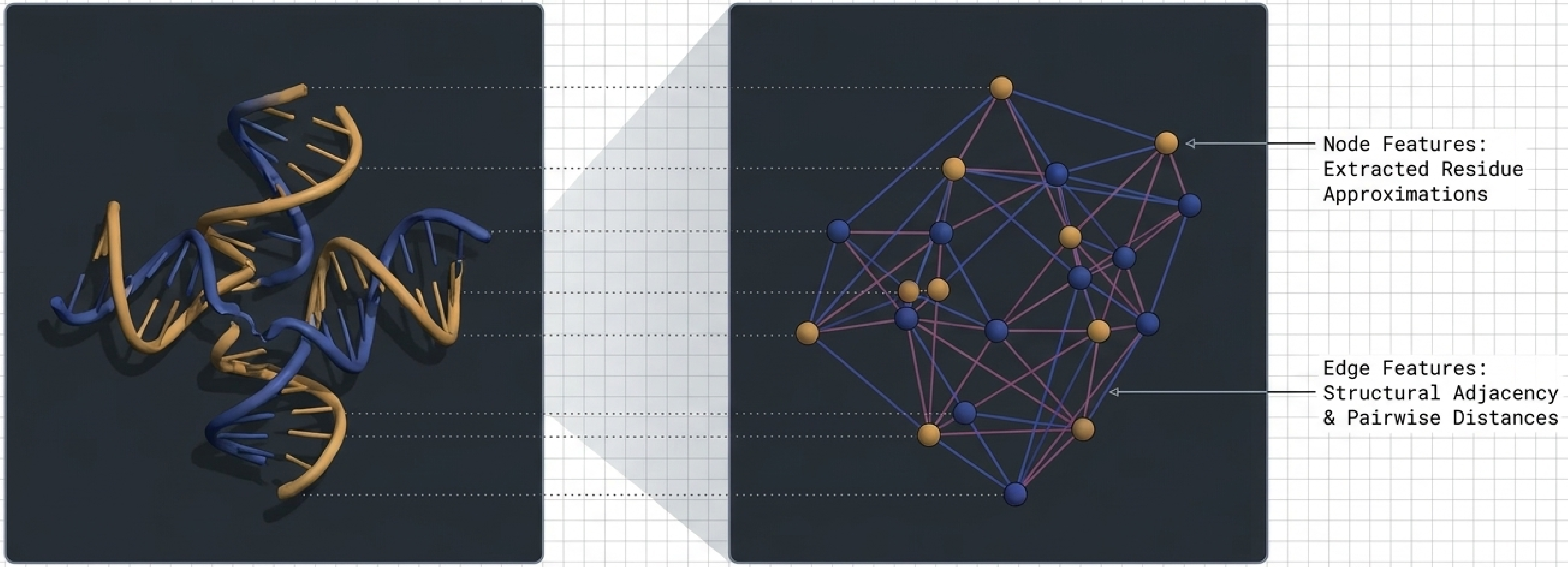
Deep Dive: Kinetic Frustration: Sequences are explicitly engineered with decoy interactions. ML models must learn to differentiate between realistic bonding patterns and highly probable but unstable decoys to predict the true native state.

The Cross-Modal Alignment



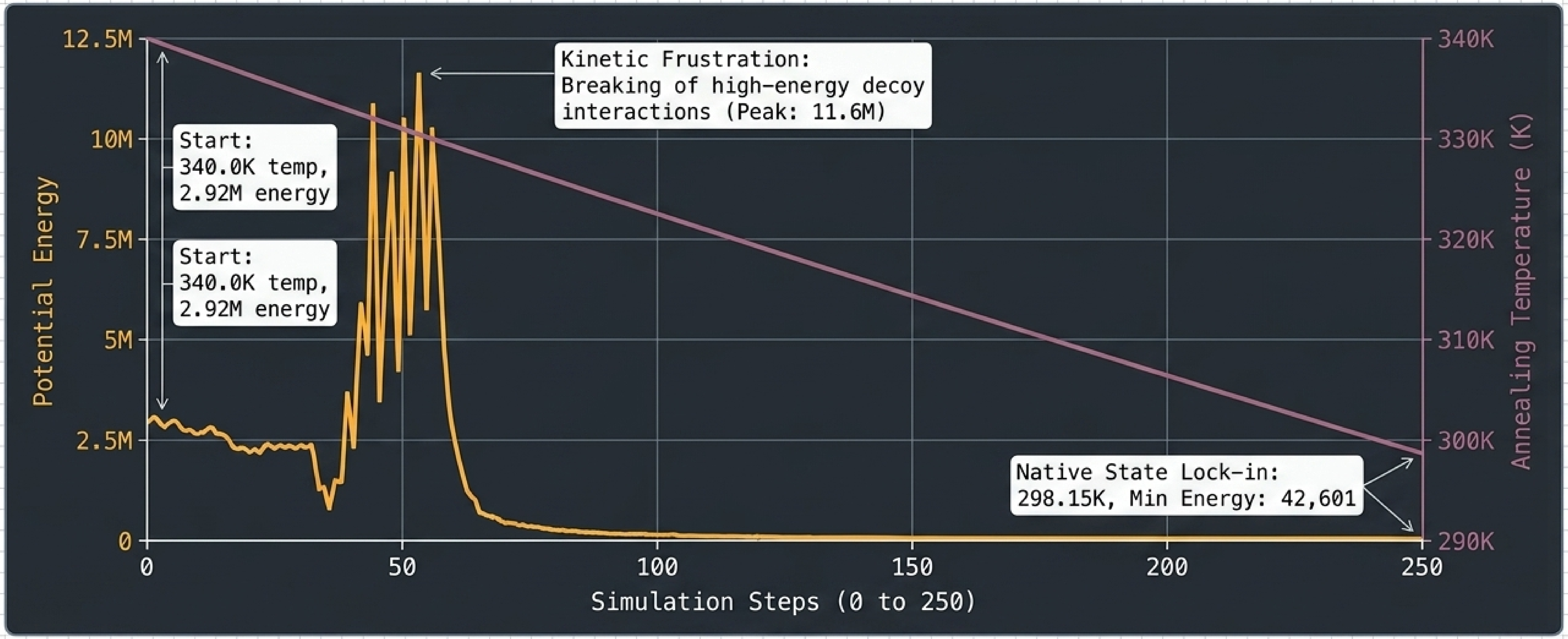
The power of SynthForge-RNA3D lies in perfectly synchronized cross-modal ground truth. A single physical folding event simultaneously yields aligned sequences, graphs, volumes, and signals—enabling true multimodal AI architectures.

PyTorch Geometric (PyG) Extraction



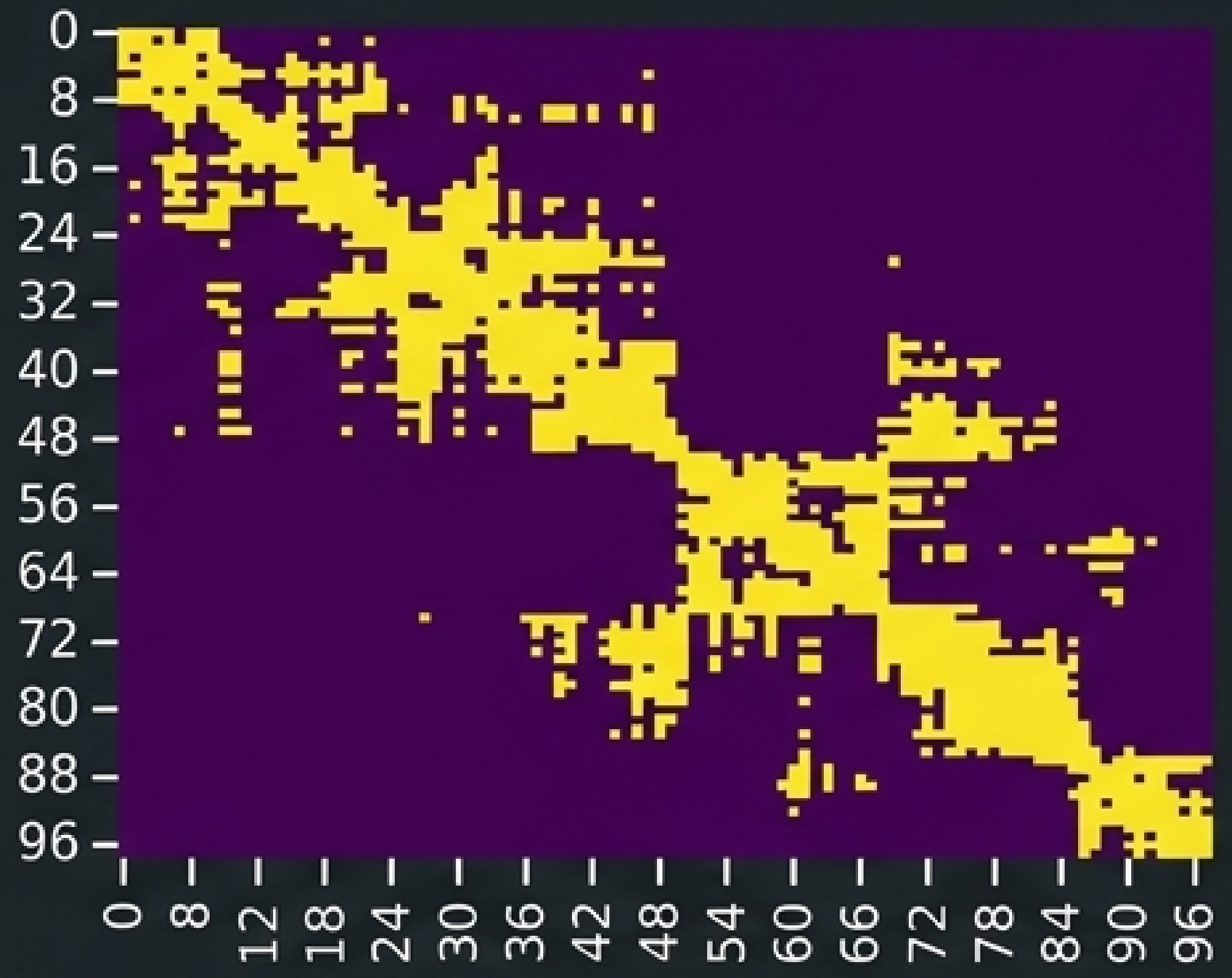
Deep Dive:
SE(3)-Equivariance: By exporting direct distance and adjacency graphs alongside mmCIF atomic coordinate approximations, SynthForge provides native, pre-processed environments optimized for message-passing systems and graph transformers.

The Thermodynamic Annealing Funnel



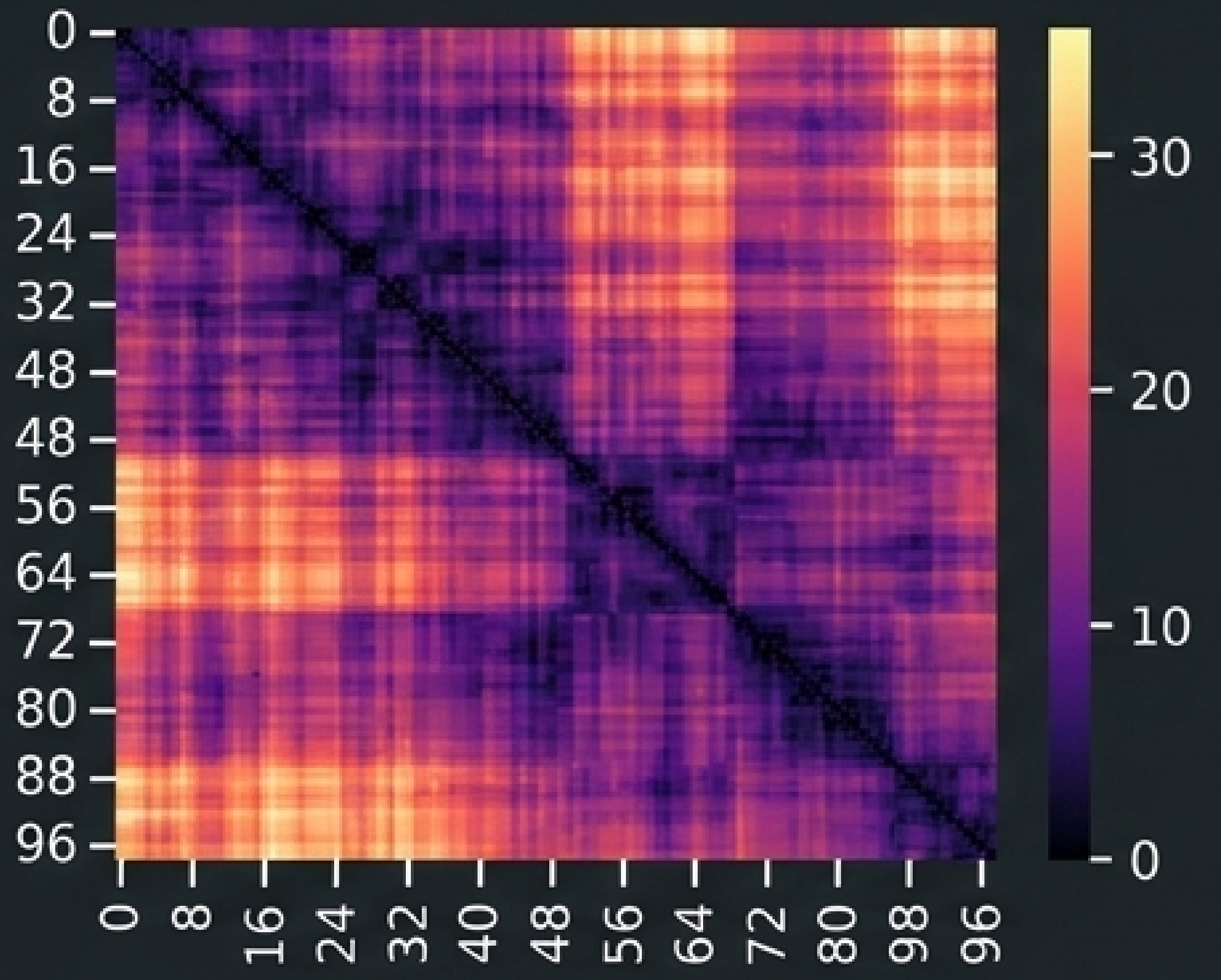
Simulated Annealing: High initial heat (340K) breaks early decoys. As temperature smoothly drops, the algorithm forces convergence toward the true global energy minimum.

2D Pairwise Geometry and Contact Mapping



Native Contact Map

Boolean contact networks for spatial adjacency learning.

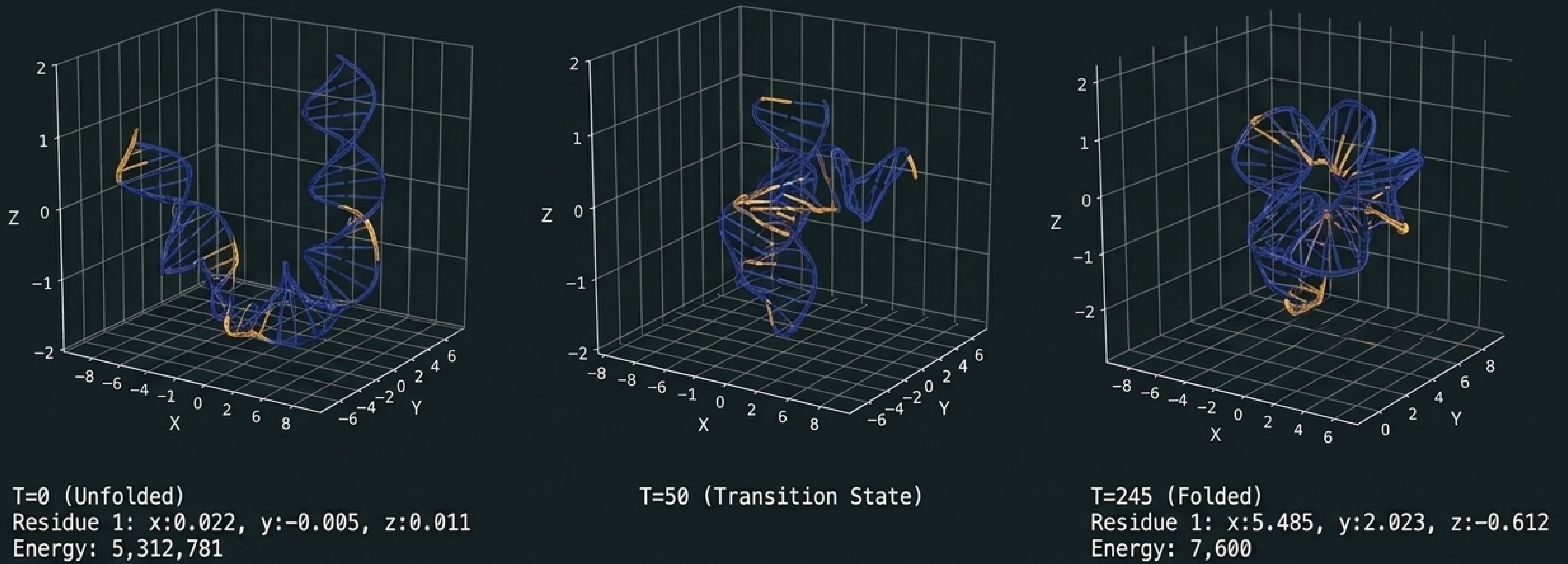


Pairwise Distance Matrix

High-resolution pairwise distance targets (0–30Å).

Deep Dive:
Dimensional Reduction: By flattening 3D constraints into 2D matrices, the dataset allows standard CNNs and 2D-attention mechanisms to learn long-range spatial dependencies without processing full 3D volumetric grids.

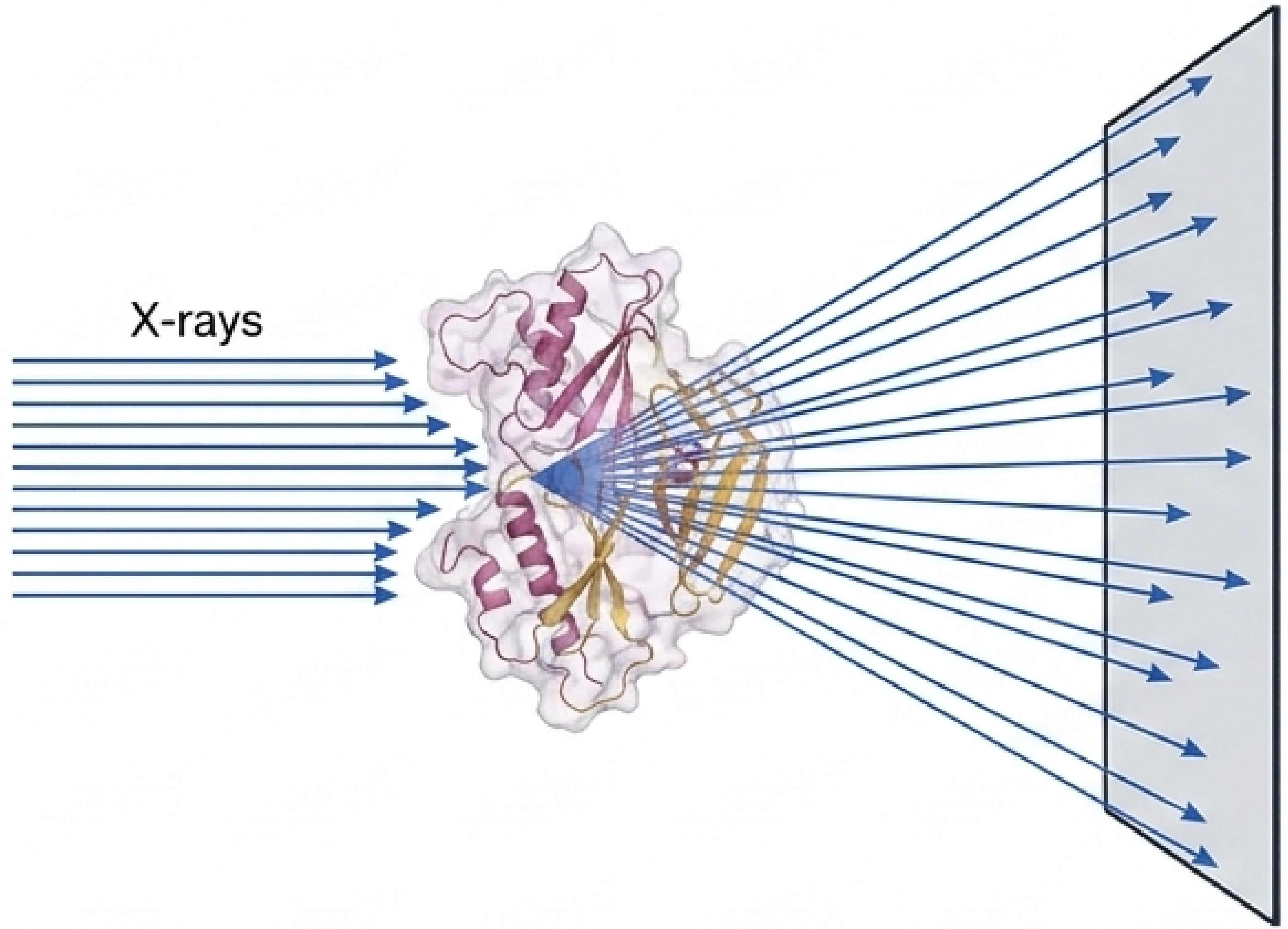
Time-Resolved Coordinate Evolution

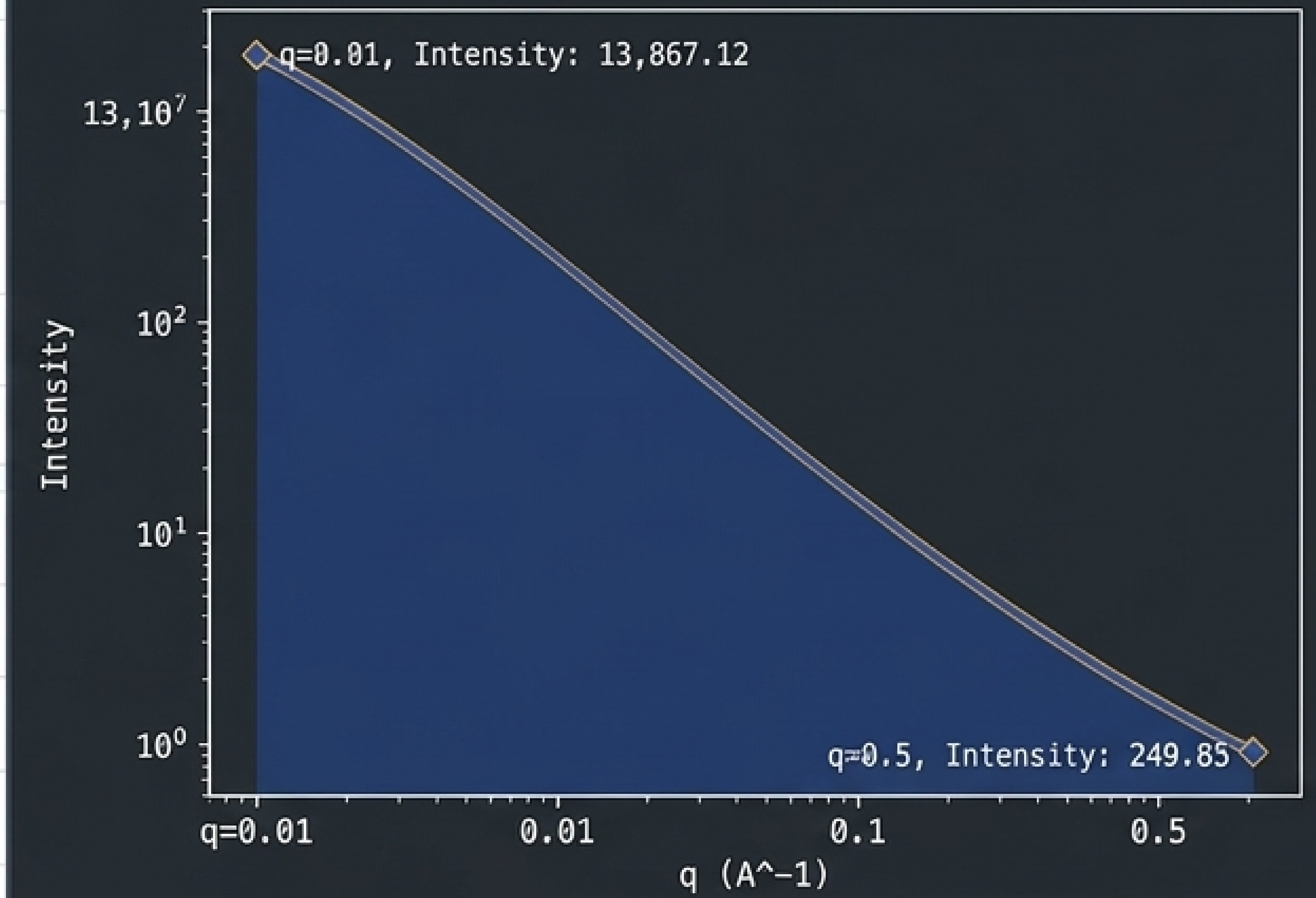


Key Insight:
Beyond the Endpoint. Standard datasets provide only the final folded state. SynthForge records the complete pathway, enabling ML models to learn *how* RNA folds, crucial for molecular trajectory forecasting and dynamic conformation generation.

Simulated SAXS Profiles for Inverse AI

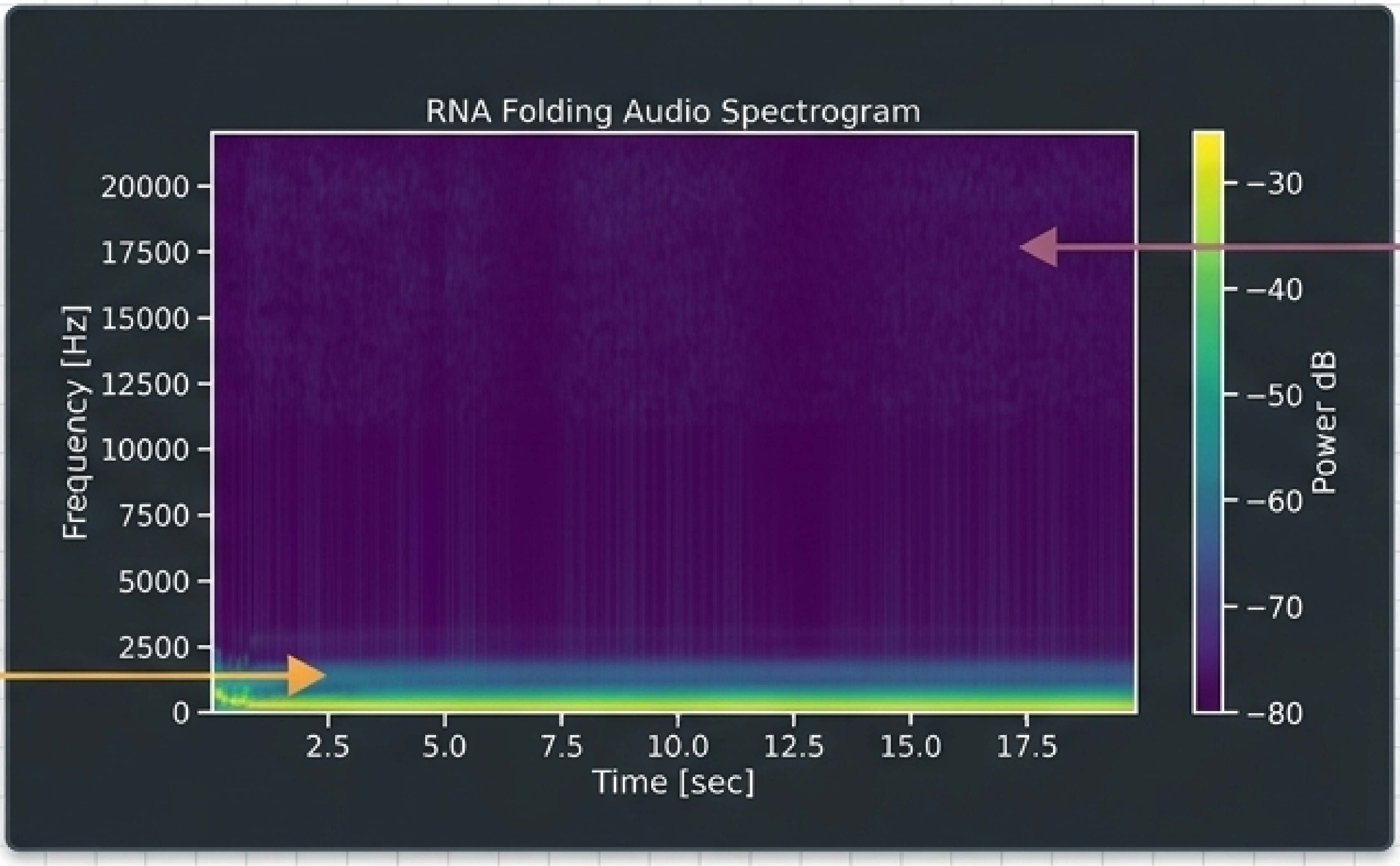
1D Scattering Signal Generation





The SAXS Inverse Problem: AI models can be trained to reconstruct 3D volumes purely from 1D scattering intensity curves. SynthForge provides paired ground truth—the exact simulated scattering curve synchronized with the 3D geometry that produced it.

Spectral Signatures & Sonification



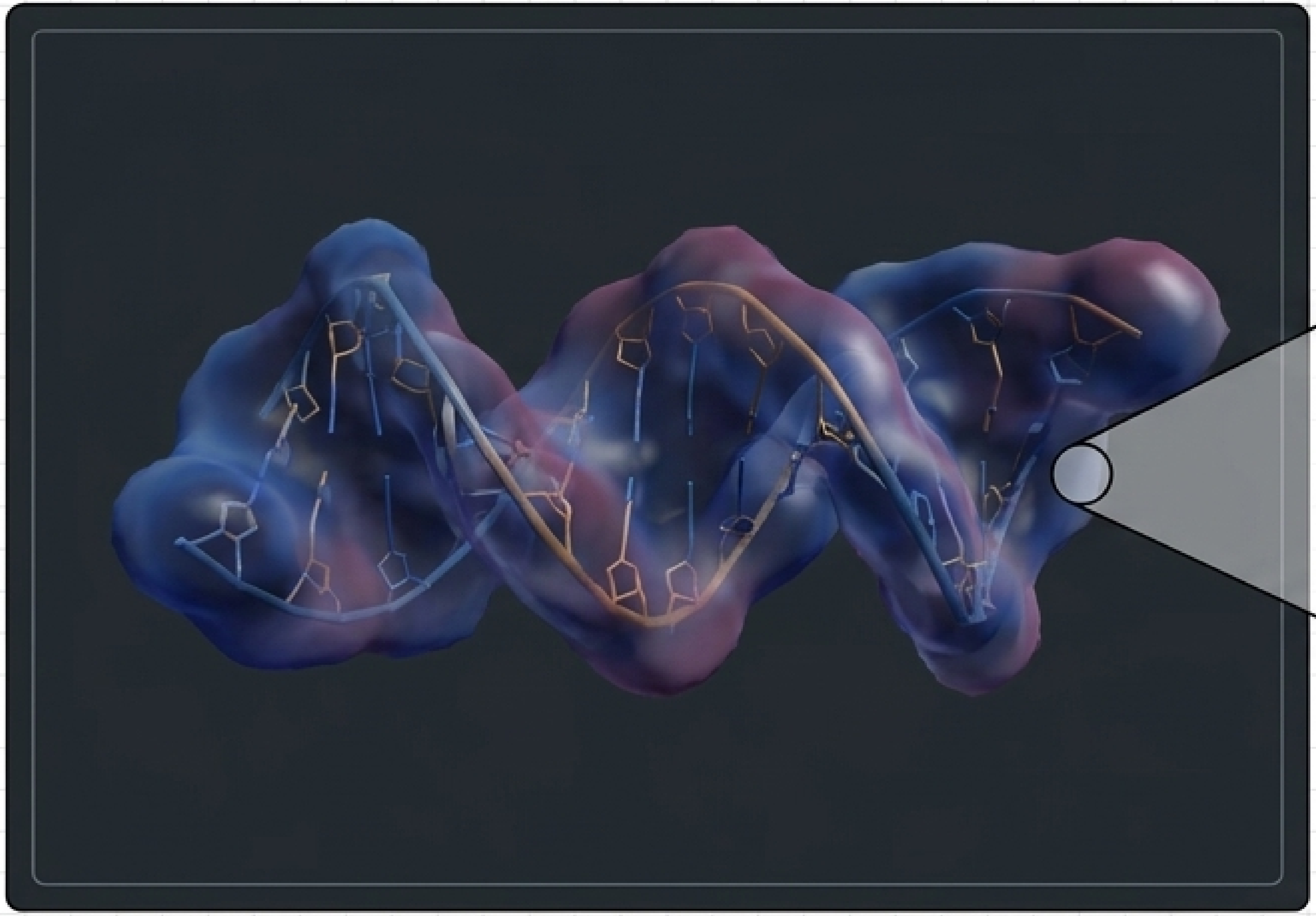
Low-frequency structural backbone stabilization.

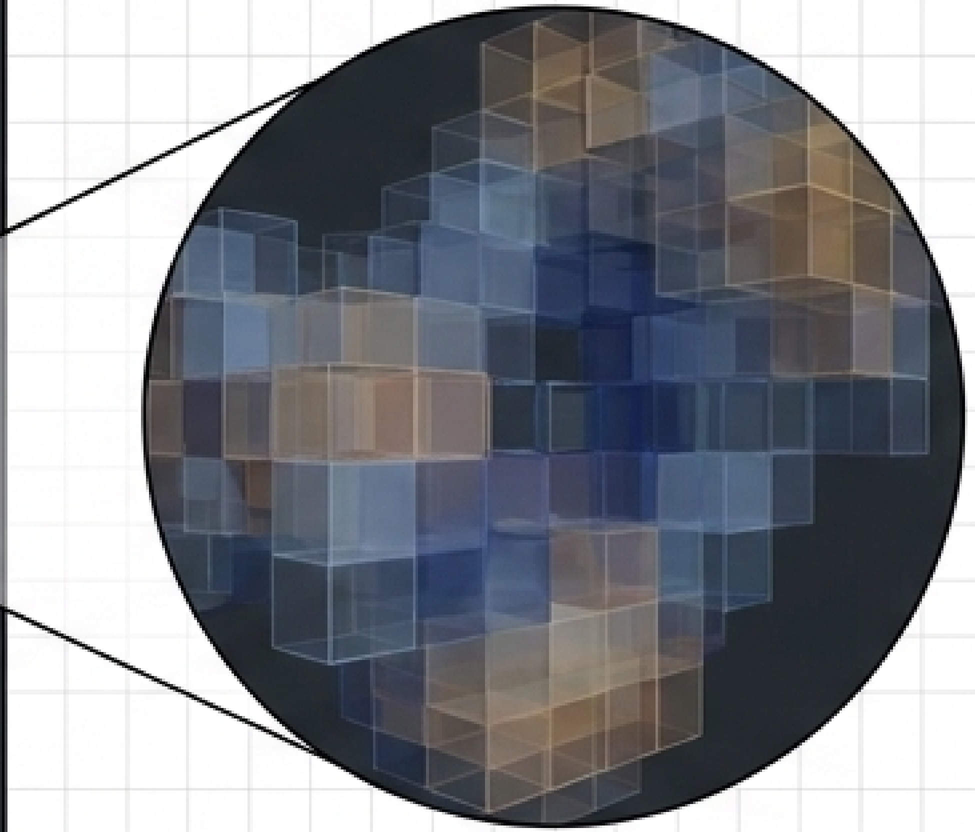
High-frequency thermal noise and localized energy fluctuations.

Deep Dive

Acoustic Mapping: Time-resolved structural shifts are mapped to audio spectrograms. This novel modality allows the application of advanced audio-based foundation models (e.g., Wav2Vec architectures) to biological folding pathways.

Synthetic CryoEM Density Grids





Volumetric density grids
& Gaussian-blurred
electron density

Deep Dive

CryoEM-Aware Learning: Real-world CryoEM data is notoriously noisy. SynthForge provides pristine, physics-based synthetic density maps, creating an ideal ground truth environment for training 3D denoising networks and structure-conditioned reconstruction models.

Ground Truth Metadata Validation

TARGET SAMPLE: SYN_RNA_00008	
Sequence Data:	UACGUGGGCAACUCUUGGCGUCUCAUGC... (Length: 75)
Interaction Profile:	Native Pairs: 18 | Decoy Pairs: 25
Ionic Environment:	Mg^{2+}: 4.64 mM | Na^{+}: 139.77 mM | Debye Length: 10.80 Å
Thermodynamic Profile:	ΔG: -7.859 kcal/mol | ΔH: -114.0 | ΔS: -0.356
Melting Temp (T_m):	593.37 K

Key Insight:
Physical Stability Mapping. Every sequence is rigorously mapped to its specific ionic conditions and energy variables, providing explicit physical constraints for conditional diffusion models.

Mapping Modalities to AI Applications

Generative Bio-AI

Input Modality:
3D Trajectories, Thermodynamic CSVs.

Application:
RNA diffusion models, structure-conditioned generation, latent space molecular modeling.

Structural Biology AI

Input Modality:
Pairwise Geometry, mmCIF.

Application:
3D structure endpoint prediction, conformation generation, folding pathway analysis.

Specialized Signal Processing

Input Modality:
SAXS Profiles, CryoEM Densities.

Application:
CryoEM denoising, 1D-to-3D SAXS shape derivation.

Dataset Specifications & Access

USABILITY SCORE: 9.41 (Kaggle Metric)

UPDATE FREQUENCY: Daily

ENGINE VERSION: v6.0

LICENSE: MIT

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├── /fasta/
├── /cif/
├── /pyg_graphs/
├── /energy_logs/
└── /saxs_profiles/
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A robust, high-fidelity alternative for Bio-AI pretraining where real-world atomistic data is scarce.